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# Beyond the Laptop: Scaling Secure Dev Environments with Nomad and Consul Mesh

TLK 1335

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# The problems AI amplifies

## Code on laptops

The agent sends it to a model provider on every prompt.

## Public registries

The agent installs packages it hallucinated, or that were hijacked.

## Credentials on disk

The agent uses them for every command, unattended.

## Open egress

The agent can curl, git push or pip install anywhere.

## Unmetered API keys

The agent spends without a ceiling.

## Access to Production Resources

Many networks are not perfectly segmented. Many organizations restore production data into their dev/test databases.

# What Remote Development Workspaces is

The laptop keeps an SSH key and an editor. Everything else stays in the datacenter.

## The laptop

* VS Code or a terminal
* One SSH key per workspace
* No source, no secrets

## Three doors in

* Traefik edge port per workspace
* SSH jump bastion
* Portal behind Ory Kratos

## Workspace

* Devcontainer, UID 1001
* Persistent host volume
* Companion Service Container(s)

## Three doors out

* Squid egress proxy: domain allow-list
* SSH egress: allowed hosts
* LiteLLM: key and budget

# The HashiStack is the security boundary

At the edges:
Traefik, Ory Kratos, Nexus, Squid, LiteLLM, GitLab CE

## Nomad: tenancy

* A namespace per developer
* Workload identity for every task
* Podman driver, non-root containers
* Dynamic host volumes
* Node pools for placement

## Consul: perimeter

* Connect mTLS between services
* Intentions are the allow-list
* Terminating gateways are the doors out
* KV scoped to nomad/<user>/

## Vault: identity

* Workload JWTs bound to the nomad_namespace claim
* Human logins become namespace-scoped tokens
* SSH keys born in Vault, templated at start

## Terraform: access

* Every grant is one line in a tracked file
* Reviewable in a pull request
* Per-datacenter state

# What the agent can no longer do

| The agent tries to… | What happens |
|---|---|
| send code to a model provider directly | refused by the proxy; the only model route is litellm.virtual.consul |
| spend without limit | a per-workspace virtual key with a budget and a model list |
| install a hijacked package (ctx, event-stream) | refused by a Nexus routing rule; nothing leaves the datacenter |
| curl a host nobody approved | 403 from the proxy, in its log |
| push over SSH to a host outside its list | "administratively prohibited" at the egress bastion |
| read another developer's secrets or jobs | 403: the namespace token covers one namespace |
| become root, or touch the host | UID 1001 in a container; no route to the Nomad client host |
| deploy a DevContainer image nobody built | the portal offers only pipeline-published revisions |

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# Roadmap

## QEMU microVM Support

* Kernel-level isolation
* Fedora Minimal VMs for developer workspaces
* Guix VMs for SSHadowForge runners

## SSHadowForge

Supplant use of GitLab with SSHadowForge

* Event driven
* Hermetic builds
* Vendored dependencies
* Complete provenance attestation set and cryptographic signatures
* Reproducible builds

## Simplified Day 2 Operations

* CUE configuration source

# Live demo: a developer and an agent, inside the boundary

01 Connect VSCode

02 Connect Terminal

03 Connect local db client

04 The agent works

05 The agent hits walls



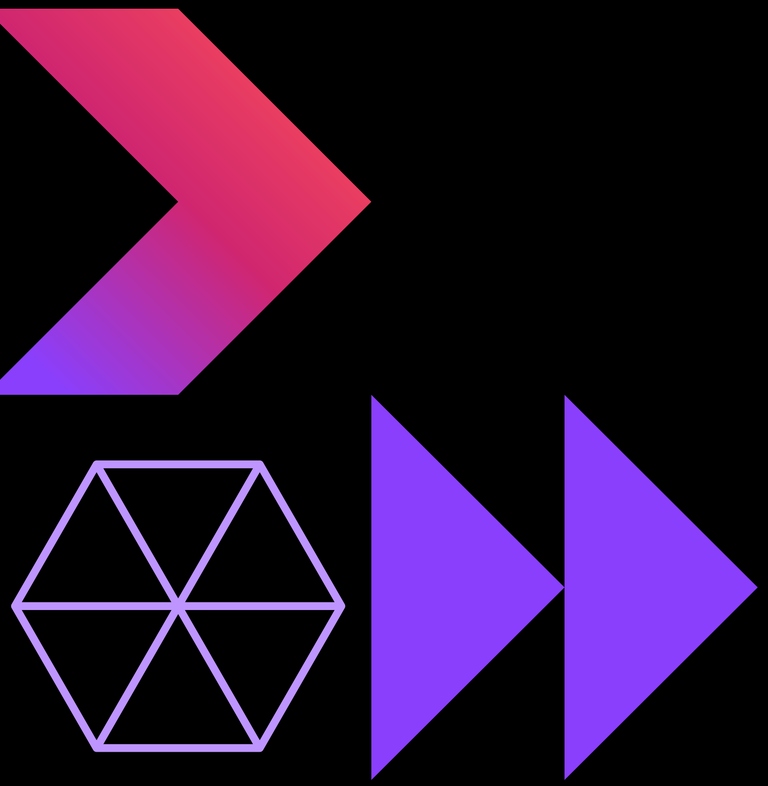
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Thank you

# Q&A



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